

Air Drawing Project – 30 Days Complete Roadmap (Python + ML)

■ Tools & Requirements (Before Starting)

You will need:

- Python (latest version)
- OpenCV
- MediaPipe (for hand tracking)
- Numpy
- A laptop/PC with webcam
- (Optional for ML improvement)
- TensorFlow or PyTorch

Week 1: Basics & Environment Setup

Day 1:

- Install Python
- Install required libraries: `pip install opencv-python mediapipe numpy`
- Test webcam access with OpenCV

Day 2:

- Learn basic OpenCV: Capture video, Show frames, Draw lines and circles

Day 3:

- Learn MediaPipe hand tracking basics
- Detect hand landmarks

Day 4:

- Understand finger points (index finger tip, thumb tip, etc.)

Day 5:

- Track finger movement in real time

Day 6:

- Practice drawing dots using finger position

Day 7:

- Review and test hand detection stability

Week 2: Drawing System Development

Day 8:

- Create a blank canvas

Day 9:

- Draw continuous lines following finger movement

Day 10:

- Add color options (using gestures)

Day 11:

- Add clear screen gesture

Day 12:

- Add thickness control

Day 13:

- Improve smoothness (reduce shaking)

Day 14:

- Testing all drawing features

Week 3: Shape & Writing Recognition

Day 15:

- Detect gesture start & stop drawing

Day 16:

- Draw basic shapes (circle, rectangle)

Day 17:

- Gesture-based shape selection

Day 18:

- Collect gesture data for ML model

Day 19:

- Train simple ML model (gesture classification)

Day 20:

- Integrate ML model into app

Day 21:

- Test shape recognition accuracy

Week 4: Finalization & Documentation

Day 22:

- UI improvement (buttons, labels)

Day 23:

- Performance optimization

Day 24:

- Error handling

Day 25:

- Full system testing

Day 26:

- Bug fixing

Day 27:

- Add demo features

Day 28:

- Record demo video

Day 29:

- Write project report

Day 30:

- Final review & submission